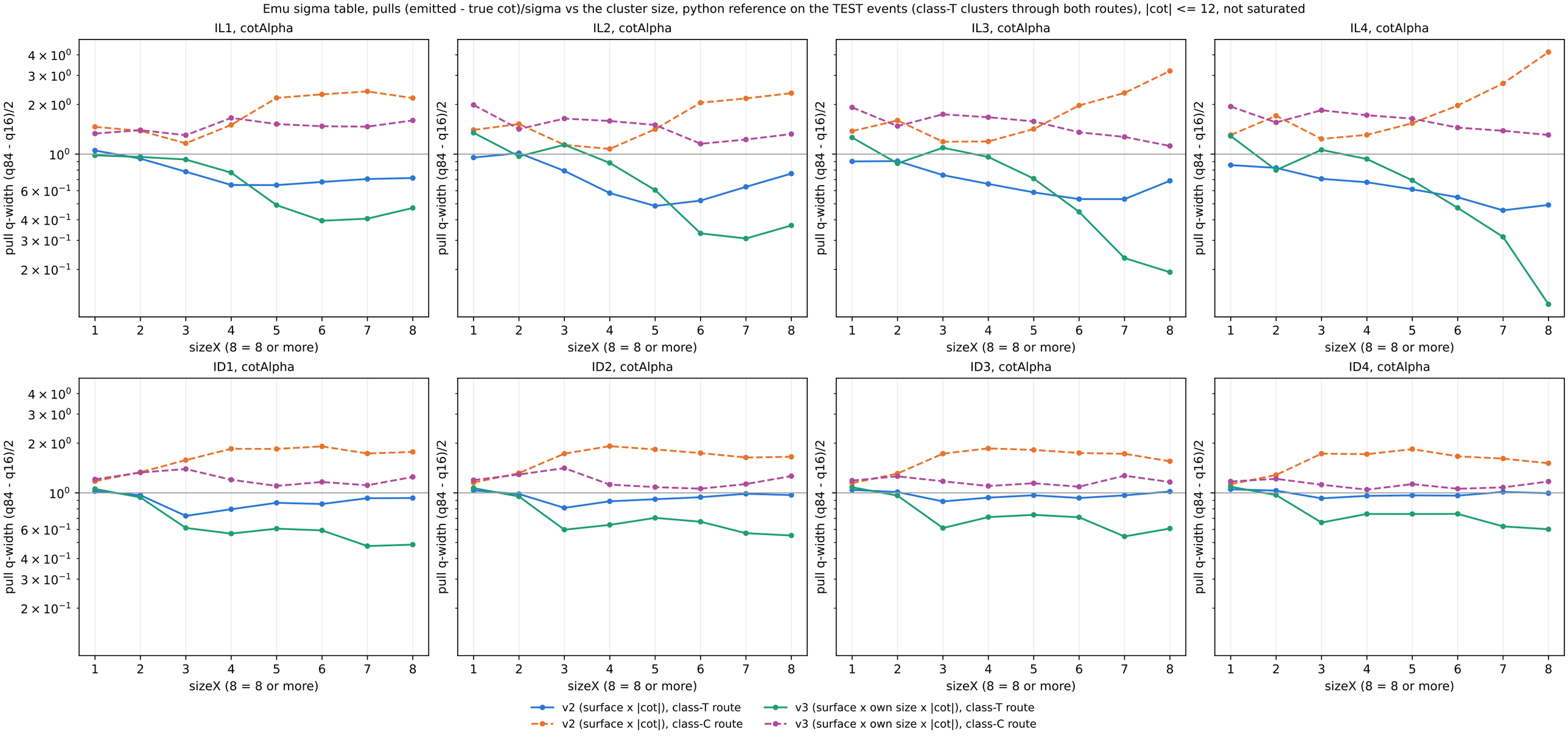
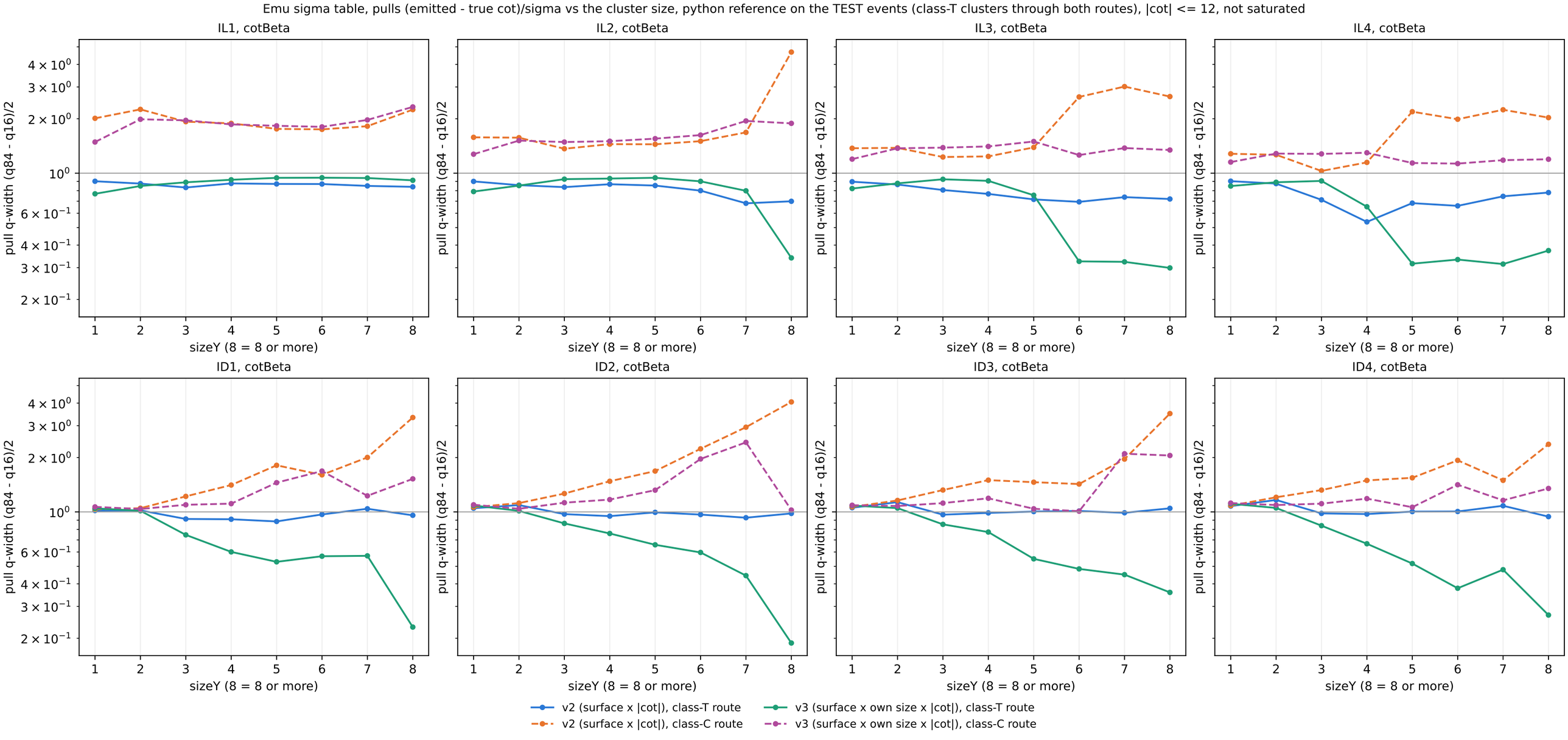
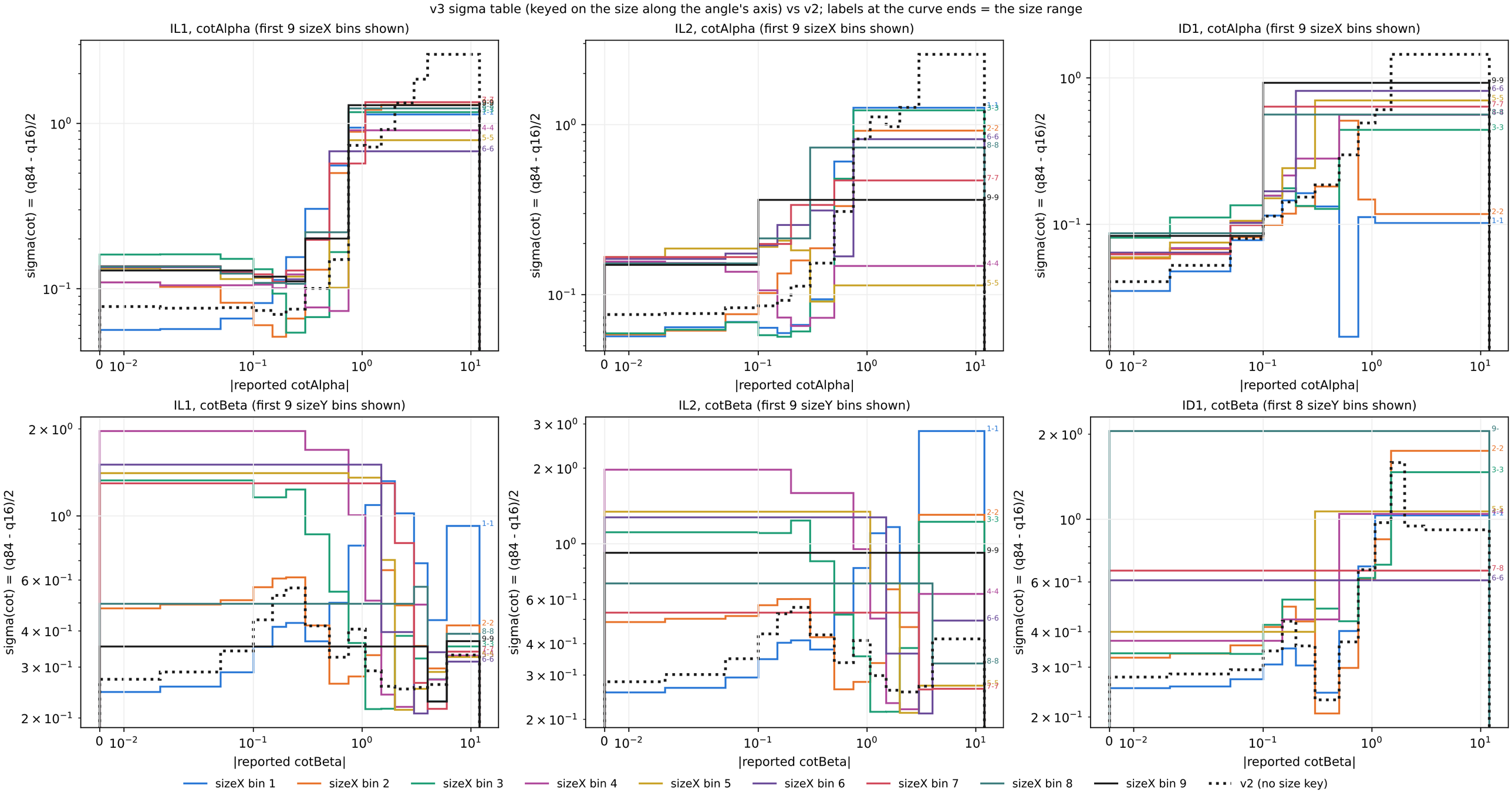

Emu sigma table, pulls (emitted - true cot)/sigma vs the cluster size, python reference on the TEST events (class-T clusters through both routes), |cot| <= 12, not saturated
IL1, cotAlpha
IL2, cotAlpha
IL3, cotAlpha
IL4, cotAlpha
ID1, cotAlpha
ID2, cotAlpha
ID3, cotAlpha
ID4, cotAlpha
pull q-width (q84 - q16)/2
sizeX (8 = 8 or more)
v2 (surface x |cot|), class-T route
v2 (surface x |cot|), class-C route
v3 (surface x own size x |cot|), class-T route
v3 (surface x own size x |cot|), class-C route


Emu sigma table, pulls (emitted - true cot)/sigma vs the cluster size, python reference on the TEST events (class-T clusters through both routes), |cot| <= 12, not saturated
IL1, cotBeta
IL2, cotBeta
IL3, cotBeta
IL4, cotBeta
ID1, cotBeta
ID2, cotBeta
ID3, cotBeta
ID4, cotBeta
pull q-width (q84 - q16)/2
sizeY (8 = 8 or more)
v2 (surface x |cot|), class-T route
v2 (surface x |cot|), class-C route
v3 (surface x own size x |cot|), class-T route
v3 (surface x own size x |cot|), class-C route


v3 sigma table (keyed on the size along the angle's axis) vs v2; labels at the curve ends = the size range
IL1, cotAlpha (first 9 sizeX bins shown)
IL2, cotAlpha (first 9 sizeX bins shown)
ID1, cotAlpha (first 9 sizeX bins shown)
IL1, cotBeta (first 9 sizeY bins shown)
IL2, cotBeta (first 9 sizeY bins shown)
ID1, cotBeta (first 8 sizeY bins shown)
sigma(cot) = (q84 - q16)/2
|reported cotAlpha|
|reported cotBeta|
sizeX bin 1
sizeX bin 2
sizeX bin 3
sizeX bin 4
sizeX bin 5
sizeX bin 6
sizeX bin 7
sizeX bin 8
sizeX bin 9
v2 (no size key)

Pulls, python reference on the TEST events (class-T clusters through both routes; 'T+C inclusive' = weighted by the test-event class fractions of each (surface, own size) cell), |cot| <= 12, not saturated: q-width and fraction beyond 3 (1)

angle	surface	route	q_width v2	q_width v3	q_width alt	frac_abs_gt3 v2	frac_abs_gt3 v3	frac_abs_gt3 alt
alpha	IL1	T	0.9149	0.9285	0.9362	0.02314	0.0129	0.01746
alpha	IL1	C	1.416	1.374	1.404	0.07082	0.05398	0.06504
alpha	IL1	T+C inclusive	1.028	1.009	1.024	0.04406	0.03111	0.0409
alpha	IL2	T	0.8788	1.031	1.045	0.008569	0.01969	0.0234
alpha	IL2	C	1.363	1.558	1.578	0.04996	0.07526	0.0806
alpha	IL2	T+C inclusive	1.03	1.149	1.171	0.03415	0.0444	0.0504
alpha	IL3	T	0.8159	0.9807	0.9924	0.01313	0.01917	0.02405
alpha	IL3	C	1.413	1.636	1.651	0.08286	0.09808	0.1013
alpha	IL3	T+C inclusive	1.035	1.149	1.17	0.06264	0.05175	0.05674
alpha	IL4	T	0.7578	0.935	0.9462	0.01341	0.01901	0.0239
alpha	IL4	C	1.452	1.715	1.721	0.1111	0.1219	0.1169
alpha	IL4	T+C inclusive	1.052	1.184	1.204	0.08726	0.06483	0.06638
alpha	ID1	T	1.002	1.006	1.012	0.007139	0.01302	0.01326
alpha	ID1	C	1.225	1.235	1.245	0.03301	0.03827	0.03616
alpha	ID1	T+C inclusive	1.085	1.053	1.059	0.03361	0.03443	0.03499
alpha	ID2	T	1.019	1.027	1.027	0.006437	0.01605	0.01581
alpha	ID2	C	1.2	1.215	1.218	0.03038	0.03883	0.0397
alpha	ID2	T+C inclusive	1.095	1.066	1.07	0.03315	0.03784	0.03822
alpha	ID3	T	1.034	1.043	1.043	0.008189	0.01849	0.01994
alpha	ID3	C	1.188	1.199	1.204	0.03048	0.03854	0.04089
alpha	ID3	T+C inclusive	1.104	1.076	1.08	0.03605	0.03846	0.04191
alpha	ID4	T	1.043	1.052	1.051	0.01963	0.02421	0.02496
alpha	ID4	C	1.167	1.178	1.182	0.04056	0.04294	0.0453
alpha	ID4	T+C inclusive	1.112	1.081	1.084	0.0506	0.04407	0.04851
beta	IL1	T	0.8665	0.8767	0.8864	0.0154	0.007809	0.009695
beta	IL1	C	2.001	1.891	1.897	0.1974	0.1673	0.1618
beta	IL1	T+C inclusive	1.003	1.003	1.015	0.05427	0.03675	0.03687
beta	IL2	T	0.8574	0.8713	0.8894	0.01732	0.007551	0.009857
beta	IL2	C	1.501	1.462	1.498	0.09542	0.06579	0.06942
beta	IL2	T+C inclusive	1.01	1.003	1.028	0.04419	0.02218	0.02449
beta	IL3	T	0.8482	0.8708	0.8896	0.01542	0.007063	0.009064
beta	IL3	C	1.332	1.324	1.357	0.05594	0.04	0.04355
beta	IL3	T+C inclusive	1.012	1.007	1.029	0.03369	0.01689	0.01918
beta	IL4	T	0.8482	0.8735	0.8879	0.01314	0.006555	0.007774

Pulls, python reference on the TEST events (class-T clusters through both routes; 'T+C inclusive' = weighted by the test-event class fractions of each (surface, own size) cell), |cot| <= 12, not saturated: q-width and fraction beyond 3 (2)

angle	surface	route	q_width v2	q_width v3	q_width alt	frac_abs_gt3 v2	frac_abs_gt3 v3	frac_abs_gt3 alt
beta	IL4	C	1.227	1.235	1.26	0.03867	0.02838	0.03083
beta	IL4	T+C inclusive	1.012	1.005	1.021	0.02725	0.01444	0.01624
beta	ID1	T	1.017	1.029	1.027	0.007132	0.007686	0.008797
beta	ID1	C	1.041	1.055	1.06	0.01153	0.01122	0.01235
beta	ID1	T+C inclusive	1.031	1.036	1.037	0.01527	0.01367	0.01528
beta	ID2	T	1.058	1.054	1.048	0.001739	0.003086	0.004001
beta	ID2	C	1.079	1.076	1.073	0.006927	0.006628	0.007409
beta	ID2	T+C inclusive	1.076	1.061	1.056	0.01126	0.009613	0.01099
beta	ID3	T	1.07	1.072	1.071	0.0005765	0.0006645	0.001373
beta	ID3	C	1.087	1.086	1.086	0.005727	0.004598	0.004852
beta	ID3	T+C inclusive	1.088	1.078	1.077	0.009303	0.007547	0.008269
beta	ID4	T	1.093	1.094	1.091	0.002673	0.003693	0.003734
beta	ID4	C	1.107	1.114	1.112	0.009015	0.00924	0.008801
beta	ID4	T+C inclusive	1.11	1.101	1.098	0.01204	0.01109	0.01151

Summary over the 8 surfaces x 2 angles: mean / max |log q-width| (0 = ideal) and the fraction beyond 3

table	route	mean_abs_log_qwidth	max_abs_log_qwidth	mean_frac_gt3	max_frac_gt3
alt (surface x sizeX x sizeY x |cot|)	C	0.2817	0.6404	0.05406	0.1618
alt (surface x sizeX x sizeY x |cot|)	T	0.0629	0.1206	0.01357	0.02496
alt (surface x sizeX x sizeY x |cot|)	T+C inclusive	0.07228	0.1854	0.03255	0.06638
v2 (surface x |cot|)	C	0.2488	0.6935	0.05436	0.1974
v2 (surface x |cot|)	T	0.1032	0.2774	0.01082	0.02314
v2 (surface x |cot|)	T+C inclusive	0.05312	0.1062	0.0368	0.08726
v3 (surface x own size x |cot|)	C	0.2732	0.6369	0.05256	0.1673
v3 (surface x own size x |cot|)	T	0.06881	0.1384	0.01166	0.02421
v3 (surface x own size x |cot|)	T+C inclusive	0.06276	0.1688	0.02994	0.06483

Pulls in CMSSW (nano, file 3, 100 events), class T (TRUTH-ONLY split), |cot| <= 12, not saturated: v2 vs v3 table

angle	surface	q_width v2	q_width v3	frac_abs_gt3 v2	frac_abs_gt3 v3	median v2	median v3
alpha	IL1	0.9139	0.929	0.023	0.01306	-0.0184	-0.01921
alpha	IL2	0.8749	1.029	0.008389	0.01949	-0.01415	-0.01732
alpha	IL3	0.8168	0.979	0.01287	0.01947	-0.02278	-0.02688
alpha	IL4	0.7612	0.9354	0.01352	0.01846	-0.02697	-0.0295
alpha	ID1	1.002	1.007	0.007258	0.01294	-0.08825	-0.08846
alpha	ID2	1.015	1.024	0.006568	0.01597	-0.1334	-0.1304
alpha	ID3	1.034	1.044	0.008266	0.01835	-0.2174	-0.2084
alpha	ID4	1.042	1.052	0.01947	0.02387	-0.294	-0.2806
beta	IL1	0.8656	0.877	0.01562	0.007546	-0.0134	-0.01361
beta	IL2	0.8553	0.8702	0.01743	0.007585	-0.02152	-0.02224
beta	IL3	0.851	0.8704	0.01562	0.007302	-0.02723	-0.02989
beta	IL4	0.8488	0.8742	0.01342	0.006868	-0.02558	-0.02965
beta	ID1	1.017	1.028	0.007057	0.007611	0.007616	0.00831
beta	ID2	1.056	1.052	0.001833	0.00308	0.02653	0.02627
beta	ID3	1.07	1.073	0.000572	0.0006499	0.1158	0.1189
beta	ID4	1.092	1.093	0.00265	0.003829	0.2163	0.224

Checks

check	result
new code + v2 table vs fix-round nano, SP branches	{"n_common_events": 100, "n_branches_compared": 245, "n_identical": 245, "differing": [], "only_in_one": []}
v3 vs v2 table, SP branches	{"n_common_events": 100, "n_branches_compared": 245, "n_identical": 87, "differing": ["nL1TSmartPixelsExtRefitHitDigiRefitAAAA", "L1TSmartPixelsExtRefitHitDigiRefitAAAA_layer", "L1TSmartPixelsExtRefitHitDigiRefitAAAA_hitAccepted", "L1TSmartPixelsExtRefitHitDigiRefitAAAA_windowTruncated", "L1TSmartPi
v3 vs v2 table, non-SP branches	{"n_common_events": 100, "n_branches_compared": 617, "n_identical": 617, "differing": [], "only_in_one": []}
emitted cot identical	true